Philosophy 415: Question Set #1

Due on Wednesday, September 25, 2019

University of New Mexico 2:00 PM

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Please explain why Frege's quantificational logic is better than Aristotle's syllogistic logic for treating inferences involving each of the following sentences and claims:

a. Socrates is older than Plato. (issue with more than one subject in sentence)

b. Every even number greater than 2 is the sum of two primes. (Goldbach's conjecture)

c. "Each [self-consciousness] aims at the death and destruction of the other." (cf. Hegel, *Phenomenology of Spirit*, section 187.) (Issue with model "each")

Aristotle's Term Logic (also known as syllogistic logic) was based on the subject predicate sentence structure. These sentences are known as propositions and are built from two terms. The first is the subject and has no true or false value. The second is the predicate which has a true or false value. From this, the second term should show that the predicate is affirmed or denied by the subject. These propositions can then be placed into a syllogism (If A then B, If B then C, If A then C) in which an inference is made from the propositions (creating syllogistic logic).
Moreover, Aristotle assigns propositions a universal or particular quantity and a negative or affirmative value. Combining these four qualities together creates four different types of propositions (A-Universal and Affirmative, E- Universal and Negative, I-Particular Affirmative, and O-Particular and Negative). These propositions contain an exclusive group of quantificational modifiers known as: "no", "some", or "all".
In 1870, Gottlob Frege (Mathematician, Logician, and Philosopher) discovered a limitation to Aristotle's logic (and all others) which led to the formation of his own formal symbolic logic system called *Begriffsschrift*. His new system was structured around concept-object sentence structure rather than subject-predicate. For instance, if we take the sentence "Socrates is older than Plato" and use Aristotle's Logic, then a problem occurs because of the two subjects "Socrates" and "Plato". It may seem that the subject is "Socrates" and the predicate is "is older than Plato"; however, this is not the case because the sentence is showing a relationship between Socrates and Plato. If we use Frege's logic, then we can use a predicate that takes two places in order to show the relationship between Socrates and Plato written like: $O(sp)$.
Furthermore, this shows Aristotle's logic does not function well under propositions with relations. Frege claimed that sentences created with concepts and objects function like functions. This means that the arity of the sentence-function can hold more than one argument/concept. For instance, if we take the sentence, "Every even number greater than 2 is the sum of two primes" (Goldbach's conjecture), then using Frege's logic we can take the concepts "every even number greater than 2" and "the sum of two primes" and express the output between them as relational using predicate logic (unsure of the exact notation). Thus, showing that we are able to show symbolically the relation between concepts and their objects. Aristotle's logic would have a hard time computing this statement because its arity is only limited to only one where that one argument does something.
Moreover, Aristotle's logic is also limited in its quantificational modifiers (no, some, and all). This is problem because natural language and reasoning have more modifiers than just these three. For instance, take the sentence: "Each [self-consciousness] aims at the death and destruction of the other." (cf. Hegel, *Phenomenology of Spirit*, section 187.) which has the word "each" to mean EACH self-consciousness preforms the action (not necessarily the same preformed action) of aiming at the death and destruction of the other. Therefore, we have to add the quantifier "each" to every self-consciousness which is something Aristotle's logic could not do.
Overall, Frege's Quantificational logic is better than Aristotle's Syllogistic logic when dealing with propositions that contain relations and multiple quantifications. Frege switched subject-predicate structured propositions to concept-object structured proposition. By doing so, Frege enabled his logical system to account for far more complicated propositions that Aristotle's Syllogistic logic could not compute.

2

In 1900, with reference to Leibniz, Bertrand Russell wrote, "The view that a subject and a predicate are to be found in every proposition is a very ancient and respectable doctrine; it has, moreover, by no means lost its hold on philosophy..." (*The Philosophy of Leibniz*, p. 12). However, Russell says, there are many propositions, especially in mathematics, that are not reducible to the subject-predicate form. Examples of these, according to Russell, include propositions asserting "relations of position, of greater and less, of whole and part." Explain why these relational propositions are not generally reducible to subject-predicate form and how Frege's theory of concepts as *functions* improves upon this traditional conception.
Relational propositions cannot be reduced to a subject-predicate form ($s - p$ form) because it does not give a clear conclusion of the statement. Instead the proposition breaks into two parts where one is the subject and the other is the predicate where one holds a property. However, if it's a relational proposition, then it must follow that a property is held by two or more things which share a relationship with one another. Frege's logic lays out the relationship between two or more properties by treating the sentence as a concept-object (functional) structure instead. Frege's functional sentence structure improves upon the subject-predicate structure by formulating relational propositions that hold the logically correct/intended meaning and returning a true or false value.
Given the statement 2 is greater than 1, we can deduce that both 1 and 2 hold a value such that one value is less than, greater than, or equal to the other. If we expressed this in $s - p$ form, then we would have one subject being "2" and one predicate being "greater than 1". We know this is not a correct analysis because both 1 and 2 hold the property of a value and are related by how much value one holds over the other. We also don't know what this sentence is saying by just telling us which is the subject, and which is the predicate. We really want to confirm or deny the statement so we can build arguments off true sentences. Therefore, if we used this method for analyzing the sentence, we would lose its intended meaning.
Let's try looking at the sentence as a function where we have two inputs and those inputs will return one output with either a true or false value. For instance, if we say "2 is greater than 1", then we can write: (if $2 > 1$, then return true) and (if $2 < 1$, then return false). We know intuitively that 2 is greater than 1, so we know it will return a true value whereas 1 is greater than 2 is not true, so it will return a false value. By looking at these sentences as a function, we are able to deduce one value instead of many. It's important to have one value over many because we can then use these values to build new ideas.
Overall, Frege's creation of the functional sentence structure allows relational propositions to return a truth value. Truth values help build arguments and allow for an easier way to connect new ideas. For instance, if we have a decision tree, it would be easier and faster to traverse through it with one value rather than multiple. This way of categorizing sentences improves the system because it lets us build stronger claims and propositions leading to more correct/true discoveries.

3

In his *Essay Concerning Human Understanding*, John Locke argues that the idea of an actually completed infinite is incoherent. For, Locke argues, supposing anyone has such an idea, "It would, I think, be enough to destroy any such positive idea of infinite, to ask him that has it, — whether he could add to it or no; which would easily show the mistake of such a positive idea." (Book II, Chapter 17, sect. 13) Locke also holds that "[If] a man had a positive idea of infinite ..., he could add two infinities together; nay, make one infinite infinitely bigger than another, absurdities too gross to be confuted!" (Book II, Chapter 17, sect. 20). Explain why the reasoning behind both quotes is wrong, according to Cantor's theory of sets.
First, the infinite is not an actual number, but rather a concept. So, to add one to a concept would be pointless since it is unnecessary to add a number to a concept. Instead, you would just be returned with the concept. However, this may not be a sufficient enough explanation so we will use Hilbert's hotel and some of Cantor's Theorems to explain this in more depth.

Let's assume we have a hotel with infinitely many rooms and a no vacancy sign on the door, can a new customer find a room? Yes, because if there are infinitely many rooms, then we can just move each guest over to the next room and fit the new guest at the start of the rooms. This same concept can be applied by adding one to infinity because we can just move every number over in the number line and add the new number at the beginning. If we do this, then we can add new numbers to the infinite set and still have an infinite set.

However, we shouldn't stop there. How do we know that by adding a new number to this set of infinity isn't just going to create a bigger set than the one we started with? Well, we can use Cantor's Theorem of Transfinite Numbers to show that the cardinality of both sets is exactly the same. What does cardinality mean? Well, it explains the size of a set (as opposed to ordinal which explains placement of an element in the set). This allows us to put sets into a on a one-to-one correspondence and if they correspond, they have the same cardinality (meaning the same size). For instance, if we have to infinite sets of fruits where one is pears and the other is oranges, then we can map Pear 1, Pear 2, and Pear 3, etc. to Orange 1, Orange 2, and Orange 3, etc. Knowing this, if we had the first set of infinity (before adding 1) and the second set where we added one, then we can theoretically put them into a one-to one correspondence with each other to show they're actually the same size.

Well, what if we added two infinites together, would this set be bigger than the original infinite set? The answer is still no because they still have the same cardinality. If we added the two infinites together and then mapped them, we would still find a one-to-one correspondence. Therefore, because both sets will have a one-to-one correspondence, they are equivalent in cardinality.

4

On p. 58 of *The Foundations of Arithmetic*, Frege holds that "Number is not abstracted from things in the way that colour, weight, and hardness are, nor is it a property of things in the sense that they are." Why not? How does Frege argue that statements and judgments of number do not make assertions about the properties of any objects (individual or collective)? What kind of assertion do they make, then?

If we were to think of a number as a way that color, weight, and hardness are to a concept, then we would be thinking of numbers as an adjective applied to something. However, numbers do not function in this way. For instance, if we see the color of apples as red, then we are saying each apple has the color red. A number does not function the same because if we say, "we have two apples", we are not saying that each of the apples has 2. Therefore, a number is not property applied to a specific concept in the same as color, weight, and hardness.

Instead numbers are objects that assert something about a concept. What does this mean? Well, objects have descriptions (usually referred to with a name) and concepts are things that refer to being (what are). So, numbers are descriptions that let us know something about the concepts way of being. Therefore, when we say how many apples are on the tree, we are asking about the concept *apples on the tree* and how many times that concept is presenting itself in an instance.

Moreover, this shows that we are not making a judgement about a number or object, but instead we are making a judgement about a concept. However, if we are to look at in this way, then we are losing sight of the distinction between object and concept (Frege's 3^{rd} Principle) because we are asserting that the object is a property of a concept, and that is not what we want to do. We want to avoid saying that numbers are properties to things, but instead that they are specific singular objects that tell us more about the concept. Frege realized this difficulty and instead came up with a Contextual Definition instead of a giving a general definition of a number. This means that we will use a specific standard in which we are judging an object to a concept. He wants to test his standard not just with one group but possibly two or more in order to explain judgements about numbers. He does so using cardinality and uses Hume's Principle to explain this further. Hume's Principle is that the number of F's is equal to the number of G's if and only if a bijection is present

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between them (meaning a one-to-one correspondence). Therefore, this would make both sets equinumerous to each other.
But what does that do, you might ask. Well, it means that for any concept let's say G with the object/number x, we can now say that number x is an extension of concept G. An extension is just a set of things to which the extension applies. For instance, if I have all UNM teachers, then the extension of the concept *all UNM teachers* would entail any and all teachers who have taught at UNM. Because number x is an extension of concept G, we can say that for all sets that are an extension of the concept G there must exist a bijection between the extension of G and G. Therefore, the number which tells us the being of G is the extension of the concept G where a bijection is present between the extension G and G. Or in Frege's language: The Number which belongs to the concept F is the extension of the concept "equinumerous to the concept F".

5

Each of the following gives a 'definition' which may or may not pick out a set, according to the axioms of ZFC. In each case, say whether or not there is a set so defined. If the set exists according to the axioms, explain how we know it exists (with reference to the axioms); if it does not, explain why not.

a. $\{x \mid x$ is a set defined on this page$\}$

b. $\{x \mid x$ is a *f-successor ordinal*$\}$ where *f-successor* ordinal is defined as follows: A set X is a *f-successor* ordinal if (and only if) either $X = \varnothing$ or $X = S \cup S$ for some f-successor ordinal S.

c. $\{x \mid x$ is an ordinal$\}$ where *ordinal* is defined as follows: A set X is an ordinal if and only if X is a transitive set consisting only of transitive sets. (Recall that a set X is transitive if and only if: whenever $A \in B$ and $B \in X$, $A \in X$).

a. The first one does not have a set defined because of the Axiom of Regularity. The Axiom of Regularity is that for every non-empty set A contains an element that is disjoint from A. But the first set is saying: x such that x is a set defined on this page. This means that if the set is defined on this page where "this page" is set A, then the only elements in the set are the ones in A so there is no element present that is disjoint from A. Therefore, making a violation of this ZFC Axiom which means it is not a set.

b. The second one is a set because it follows all the axioms. However, let's use the Axiom Schema of Replacement to prove this. The Axiom Schema of Replacement says the axiom schema of replacement asserts that the image (image of a function is the set all output values it may take) of any set under any definable mapping is also a set. So the set is saying x such that x is an ordinal where x is an f-successor if and only if x is the empty set or x is the union between S and set S ($\{S\}$). Because we are unionizing S with the set of S, we are asserting that one, we have a subset $\{s\}$ and we have another subset set ($\{S \cup \{S\}\}$) such that x is an f-successor and the empty set is also a subset set of the original set. We are taking all of these definable sets and mapping it to the original set such that it allows us to prove the original set is set. It's a set because we have definable subsets that match to the original set.

c. The third one is also a set because **b** is considered a set. This means that since the set is saying x such that x is an ordinal where ordinal is a set X if and only if x is a transfinite set consisting of only other transfinite sets. Well, we distinguished in set be that set **b** is made up of other sets using the Axiom Schema of Replacement. Since we proved that in **b** we have sets that make up other sets, we know that we have transitive sets consisting of other sets. For instance, set $\{S\}$ is a member of set $\{S \cup \{S\}\}$ and set $\{SU\{S\}\}$, {empty set} are members of $\{x \mid x$ is a f-successor ordinal$\}$ and those sets are subsets $\{x \mid x$ is an ordinal$\}$, we have shown that we have transitive sets containing transitive sets. Therefore, we have proven **c** is a set.

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[Extra Credit]. Consider the set discussed in 5b), above. If this set exists, is it itself an *ordinal*? Is it an *f-successor ordinal*? Why or why not, in each case?

A. Yes, the set is an ordinal because all its subsets must exist, making it well-ordered.

B. Yes, by definition of an *f-successor* ordinal, it is a set that contains only other *f-successor* ordinals, making it an *f-successor* ordinal itself.